

$\mathcal{O}(n)$ alternative to Quantum Fourier Transform with efficient neural net classical post-processing

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The Quantum Fourier Transform (QFT) is required by hidden subgroup algorithms including Shor’s algorithm for factoring. The circuit depth and long-range gates of the QFT remains challenging for near-term hardware. To find more shall alternatives we identify two functional properties that are exploited by the QFT to enable HSP. Firstly the shift invariance property of the QFT allows for the removal of a random overall shift. Secondly, the QFT retains information about the hidden subgroup generator accessible in the computational basis. We quantify that information via the discrete Fisher information. We construct a family of shallow circuits using Hadamards and controlled-Phase gates, HP-L circuits, that preserve shift invariance. Numerical analysis shows these circuits retain Fisher information about the generator with the same scaling as the QFT. Numerical experiments with the $\mathcal{O}(n)$ HP-1 replacing the $\mathcal{O}(n^2)$ QFT demonstrates successful Shor factoring, using an efficiently scaling neural network for classical post-processing.

Introduction.— The $\mathcal{O}(n^2)$ circuit depth required by the standard Quantum Fourier Transform (QFT) presents experimental challenges. The QFT is used in the exponential quantum speedups of landmark protocols—such as Shor’s algorithms for prime factorization and discrete logarithms [? ?], as well as Simon’s algorithm [?]. The associated family of algorithms is known as the (quantum) hidden subgroup problem (HSP) algorithms [? ? ?]. The $\mathcal{O}(n^2)$ depth complexity of the QFT remains a critical bottleneck for implementing HSP algorithms in hardware. For a 100-qubit system ($n = 100$), the QFT demands thousands of circuit layers, which vastly exceeds current hardware capabilities. For instance, IBM’s 127-qubit Eagle processor is limited to a depth of 60 layers even with extensive error mitigation [?], and Google’s 105-qubit Willow processor reaches its operational threshold at around 40 layers [?]. Quantum error mitigation protocols improve the performance, but with a sampling overhead that scales exponentially with the circuit depth [? ?]. Strategies to mitigate the circuit depth required by the QFT have primarily focused on reducing hardware execution costs, often by only approximating the QFT or by shifting part of the computation to classical post-processing. These efforts have enabled record experimental QFT fidelities [?] through techniques including (i) truncated controlled rotations [?], (ii) optimized CNOT routing tailored for restricted hardware topologies [?], which approximates the QFT with $\mathcal{O}(n \log n)$ depth, (iii) dynamic mid-circuit measurements [?], and (iv) quantum circuit cutting, which trades coherent quantum depth for classical post-processing resources [? ?]. Despite these advances, approximation-based methods still retain $\mathcal{O}(n \log n)$ depth in representative constructions, while circuit-cutting-based protocols

shift the complexity to the classical domain and can require exponential classical resources [? ?].

We here accordingly seek to design more shallow alternatives to the QFT without introducing prohibitive classical post-processing burdens. Rather than approximate the QFT we take the approach of identifying key features of the QFT that make it compatible with the HSP. WE focus on shift-invariance combined with retention of information about the generator in question. Fig. ??(a) summarizes the main idea.

We construct a shallow circuit family, called HP circuits, that preserves shift invariance, whilst retaining information about the generator. These circuits interleave transversal Hadamard (H) gates with L layers of phase (P) blocks. The circuits are shift-invariant and numerical scaling analysis demonstrates exponentially increasing retention of Fisher information $\sim e^{kn}$, albeit with a worse exponent k than QFT. That $\sim e^{kn}$ implies efficient scaling of the number of experimental runs required.

We numerically simulate replacing the QFT in Shor’s algorithm with HP-1 circuits, and using an efficiently scaling neural network [?] for classical post-processing. Factorisation is successfully achieved.

Necessary condition for Shift Invariance.— The standard quantum HSP algorithm is illustrated in Fig. ??(b) with gray blocks [? ? ? ?]. After U_f , the state is proportional to $\sum_x |x\rangle |f(x)\rangle$. The x ’s are elements of a finite abelian group G with group addition $+$. We are guaranteed that $f(x+r) = f(x)$ for the unknown generator r that we wish to find. In other words, the function $f(\cdot)$ is constant on cosets of V , the subgroup generate by r . Upon measuring (or imagining) measuring the second register to find $|f(c)\rangle$ the state becomes proportional to $\sum_{n \in \{0,1,2,\dots\}} |c+nr\rangle |f(c)\rangle$ with the shift c



FIG. 1. **Shallow shift-invariant alternative to the QFT.** (a) Shift invariance is a critical property of the QFT. We identify a circuit family, denoted as HP- L , that inherently preserves the property. Within this broader family, HP-1 specifically serves as shallow-depth gates that are friendly to experiment. (b) In the proposed HSP algorithmic routine, the deep QFT block (the left gray gate) is directly replaced by the HP circuit. A neural network then process the measurement outcomes, bypassing the standard continued fraction expansion (CFE) typically used in standard HSP. The above figure illustrates the HP circuit, which composed by Hadamard gates and phase gates.

acting as undesired noise obfuscating r (additional details about the HSP algorithm are provided in Appendix ??).

We now define shift invariance for HSP sampling and derive a necessary condition that any shift-invariant unitary must satisfy. For a subgroup $V \leq G$, we denote the uniform superposition over its elements as $|V\rangle = \frac{1}{\sqrt{|V|}} \sum_{v \in V} |v\rangle$.

Definition 1 (Shift invariance). *A unitary U acting on the group register is shift-invariant for HSP sampling if*

$$|\langle x|U|V\rangle|^2 = |\langle x|U|c+V\rangle|^2, \quad \forall x, c \in G, \quad \forall V \leq G. \quad (1)$$

To identify shift-invariant circuits beyond the QFT, we first establish a necessary condition for shift invariance.

Proposition 1 (Necessary condition of shift invariance). *Suppose U is a unitary on the group register that satisfies the shift-invariance condition above. Then, for every $x, y \in G$ there exist phases $\theta_{xy} \in \mathbb{R}$ such that*

$$\langle x|U|y\rangle = \frac{1}{\sqrt{|G|}} e^{i\theta_{xy}}. \quad (2)$$

Proof sketch of Proposition ??. The proof hinges on analyzing the shift-invariance condition for the trivial subgroup $V = \{0\}$. For subgroup $V = \{0\}$, the state

$|V\rangle$ is the computational basis state $|0\rangle$, and the shifted state is $|c+V\rangle = |c\rangle$. The requirement (Eq.??) that $|\langle x|U|0\rangle|^2 = |\langle x|U|c\rangle|^2$ for all $x, c \in G$ implies that, for any fixed output x , the magnitude of every matrix element in the x -th row of U must be identical. The special case induce the uniform magnitude $\langle x|U|y\rangle = 1/\sqrt{|G|} e^{i\theta_{xy}}$. The full details of this argument are provided in Appendix ??.

L -phase-block Hadamard-Phase circuits (HP): A shift invariant circuit family.— Proposition ?? suggests a simple design principle for constructing shift-invariant circuits. The condition $|\langle x|U|y\rangle| = 1/\sqrt{|G|}$ requires flat amplitude magnitudes, which are naturally generated by Hadamard gates. The remaining freedom lies in the relative phases θ_{xy} , which can be adjusted by controlled phase gates ($\text{CP}(\theta) = |00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 10| + \exp(i\theta)|11\rangle\langle 11|$) without changing these magnitudes. This motivates the Hadamard-phase (HP) circuit family we shall now introduce. As illustrated in the upper subfigure of Fig. ?? (b), the HP- L circuit is structured as an alternating sequence of Hadamard gates and phase blocks. To respect shift invariance (see Theorem ?? below) we enforce three key rules (i) each qubit is acted on by a Hadamard gate exactly once, and (ii) each qubit pair is connected by at most one controlled-phase gate, (iii) each controlled phase gate must come together with 2 Hadamards, one before and one after the phase gate. Here, “once” refers to the entire L -layer circuit rather than to a single layer. Examples of valid circuits are illustrated in Fig.??

To express the allowed circuits mathematically, we define two types of sets of qubits. Let H_i denote the Hadamard gate applied to qubit $i \in \{1, 2, \dots, n\}$. For each layer l , let $\Lambda_l \subseteq \{1, 2, \dots, n\}$ denote the set of qubits on which Hadamard gates are applied in that layer. The qubits that are still allowed in terms of having a Hadamard on them in layer s after layer l , i.e. those that have not had a Hadamard on them in or before layer l , are denoted as the set $\bar{\Lambda}_{\leq l}$.

In-between the Hadamard layers the two sets are coupled via a controlled-phase block, denoted as $\text{CP}_{\Lambda_l, \bar{\Lambda}_l}(\theta_{\Lambda_l \bar{\Lambda}_l})$, which applies pairwise operations $\text{CP}_{jk}(\theta_{jk})$ between each qubit $j \in \Lambda_l$ and $k \in \bar{\Lambda}_l$. Building upon those notations, we formally define the circuit architecture:

Definition 2 (HP- L circuit). *A quantum circuit is classified as an HP- L circuit if it is constructed using the following layer-wise architecture*

$$U_{\text{HP}}^{(L)}(\theta) = \bigotimes_{i \in \Lambda_L} H_i \prod_{l=L-1}^1 \left(\text{CP}_{\Lambda_l, \bar{\Lambda}_l}(\theta_{\Lambda_l \bar{\Lambda}_l}) \bigotimes_{i \in \Lambda_l} H_i \right), \quad (3)$$

where the controlled-phase block is given by

$$\text{CP}_{\Lambda_l, \bar{\Lambda}_l}(\theta_{\Lambda_l \bar{\Lambda}_l}) = \prod_{j \in \bar{\Lambda}_l} \prod_{k \in \Lambda_l} \text{CP}_{jk}(\theta_{jk}). \quad (4)$$

The parameter set $\theta_{\Lambda_l \bar{\Lambda}_l}$ is the collection θ_{jk} , $j \in \Lambda_l$, $k \in \bar{\Lambda}_l$.

In the following, we show that HP circuits preserves shift invariance. This implies that such invariance is not unique to the QFT, but is a feature that arises in a wider family of quantum circuits.

Theorem 1. *HP-L circuits are shift invariant for $\mathbb{Z}_{2^n}$.*

Proof sketch of Theorem ??. The HP-L circuit achieves shift invariance over $\mathbb{Z}_{2^n}$ because its linear phase structure isolates the constant coset shift into an unobservable global phase. Specifically, the circuit's unitary phases are linear in the binary variables (e.g., cross-terms like $x_i y_j$). Because any subgroup over $\mathbb{Z}_{2^n}$ takes the form $V = \langle 2^p \rangle$, a coset state $|c + V\rangle$ differs from $|V\rangle$ only at fixed deterministic bit positions c_i . Due to phase linearity, these c_i contributions factorize completely into a pure phase term, rendering the final sampling distribution independent of the coset representative. The full derivation is detailed in Appendix ??.

HP-0 can be used in HSP over $\mathbb{Z}_{2^n}$.— We gives an analytical solution for HSP over $\mathbb{Z}_{2^n}$ via transversal Hadamard gates. We consider the cyclic group $\mathbb{Z}_{2^n}$ since the group can be naturally represented by computational basis. In this case, each subgroup is generated by a power of two,

$$V = \{q2^p : q = 0, \dots, 2^{n-p} - 1\} \leq \mathbb{Z}_{2^n}, \quad (5)$$

where p specifies the subgroup generator 2^p . Applying a transversal Hadamard layer to a coset state gives

$$H^{\otimes n} |c + V\rangle \propto \sum_{k \in \{0,1\}^p} (-1)^{k \cdot c} |0\rangle^{\otimes (n-p)} |k\rangle. \quad (6)$$

By measuring the state $H^{\otimes n} |c + V\rangle$, one observes that the first $n-p$ bits are always fixed to 0, while the remaining p bits are uniformly distributed over $\{0,1\}^p$. Thus, the value of p can be determined by identifying which bit positions are consistently zero. The detailed calculation and sampling algorithm are provided in Appendix ??.

HP-L for HSP over $\mathbb{Z}$: information on generator is retained.—The protocol HP-0 fails for general cyclic groups $\mathbb{Z}_q$ because the measurement distributions corresponding to different subgroups V are not distinguishable. The no distinguishability demonstrates that shift invariance alone is insufficient for successful hidden subgroup recovery, and the distinguishability should be the second condition.

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FIG. 2. Fisher-information on generator is retained by HPcircuits. The figure summarizes how the output sampling distribution of a circuit retains information about the hidden subgroup parameter. (a) Schematic of the probability computation used in the discrete Fisher information (DFI) benchmark, where the circuit maps the periodic components $\{|qr\rangle\}_q$ to output probabilities over computational-basis outcomes $|b\rangle$. (b) Average DFI versus the number of qubits for the QFT reference and several HP-family circuit settings; solid curves show the mean, with shaded bands indicating one standard deviation.

Rigorously, we use the discrete Fisher information (DFI) as a metric of the distinguishability.

$$\text{DFI}(r, n) = \sum_{x=0}^{2^n-1} \frac{(\Pr(x | r+1) - \Pr(x | r))^2}{\Pr(x | r)}, \quad (7)$$

where the x represent measurement outcome and r is the hidden subgroup generator, and $\Pr(x | r)$ is the probability of measuring x when the subgroup generator is r . The sample complexity for estimating a discrete parameter is formally governed by the DFI (Eq.(??)) via the Hammersley-Chapman-Robbins bound [? ?],

$$m \geq \frac{\ln(1 + \epsilon^{-2})}{\ln(1 + \text{DFI}(r))}, \quad (8)$$

where m is the required number of samples. The bound guarantees that retaining an exponentially growing DFI results in tractable sample complexity for the subsequent neural-network-based post processing.

For standard QFT, we can express the DFI as

$$\text{DFI}_{\text{QFT}}(r, n) \approx \frac{4\pi^2}{9} \left(\frac{2^{2n}}{r^2} - 1 \right) \sim \mathcal{O}(2^{2n}). \quad (9)$$

The calculation of Eq. (??) is provided in Appendix ??.

The Eq. (??) indicates that generator r becomes easier to

recover as the qubits number increases, which motivates us to limit r with $1 \ll r \ll 2^n$.

For HP circuits, the DFI is evaluated numerically, which is illustrated in Fig. ??(a). Specifically, for a fixed r , the circuit maps the periodic input components $\{|qr\rangle\}_q$ to a computational-basis distribution over outcomes $|b\rangle$, with the exact probabilities given by $\Pr(|b\rangle) = \left| \sum_q \langle b|U|qr\rangle \right|^2$. To evaluate the worst-case parameter sensitivity as the system scales, Fig. ??(b) illustrates the minimum DFI against the qubit number n . This worst-case metric $\text{DFI}_{\min}(n)$ is formally defined as $\min_{1 \leq r \leq \lfloor 2^{n/2} \rfloor} \text{DFI}(r, n)$. We bound the periods to $1 \leq r \leq \lfloor 2^{n/2} \rfloor$ to satisfy the condition $r \ll 2^n$.

Fig. ??(b) compares the scaling of different HPcircuits configurations, where solid curves and shaded bands represent the mean and standard deviation of $\text{DFI}_{\min}(n)$. The fixed-phase HP-1 circuit utilizes the skeleton, $\Lambda_1 = \{1, 3, 5, \dots\}$ and $\Lambda_2 = \{2, 4, 6, \dots\}$, with controlled-phase gates applying control phase gate with phase $\frac{2\pi}{2^{|i-j|}}$ between qubits i and j . The optimized HP-1 circuit adopts the same skeleton but treats the gate phases as variational parameters, trained to maximize the $\text{DFI}_{\min}$. The deeper HP- L circuits are constructed by generating valid random skeletons Λ .

The proposed HPcircuit family, particularly the shallow HP-1 configuration, reduces quantum depth requirements while preserving an exponential growth of DFI. As detailed in Appendix ??, numerical data for both the HP circuits and the QFT benchmark fit the log-linear model $\ln \text{DFI}_{\min}(n) = kn + b$.

Efficient Neural network for classical post-processing.— To bypass the exponential overhead of evaluating full probability distributions, we employ a polynomial-scaling Deep Sets neural network [? ?] to predict the hidden subgroup generator directly from the raw measurement outcomes $\{b_i\}$. The input data is generated using the optimized HP-1 circuit evaluated in Fig. ??. As depicted in Fig. ??(a), a shared feature map ϕ processes m bitstrings to compute an aggregated, permutation-invariant representation $\sum_i \phi(b_i)$. Subsequently, a beam-search decoder [? ?] performs a heuristic top- k search to rank candidate periods. By maintaining a restricted set of probable solutions, this architecture avoids exponential sample-space computations and outputs a ranked list of candidate generators for classical verification.

Replacing $\mathcal{O}(n^2)$ QFT with $\mathcal{O}(n)$ HP-1 in Shor's Algorithm.— We evaluate the optimized HP-1 circuit on Shor's algorithm, demonstrating that its shallow circuit measurements retain learnable features for successful classical post processing. Shor's Algorithm seeks the generator r satisfying $a^r \equiv 1 \pmod{N}$. We select the base a such that $1 \leq r \leq 2^{n/2}$, ensuring resolvability by the n -qubit register. As shown in Fig. ??(b), the neural

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FIG. 3. **Numerical simulation shows successful Shor factoring via HP-L and neural net post-processing.** (a) Measurement outcomes are mapped to shared features and decoded by beam search to predict the period. (b) Training curves for 9-, 10-, and 11-qubit instances show rapid improvement in recovery accuracy. (c) Shor-type factorization instances colored by the minimum number k of HP-1-based guesses needed for successful period recovery. Most instances are solved with $k = 1$, while only a few require $k = 2$ or $k = 3$, showing the broad effectiveness of HP-1-based recovery.

network achieves rapid convergence on 9-, 10-, and 11-qubit instances. Both the top-1 (correct period ranked first) and top-3 accuracies increase swiftly within a few epochs.

Our evaluation on Shor factoring demonstrates that our protocol can find the subgroup generator. As illustrated in Fig. ??(c), each composite integer is colored by the minimum number of guesses k . Most instances are solved on the first attempt ($k = 1$), indicating the raw measurement distribution provides sufficient information for period extraction.

Our evaluation under a post-circuit global depolarizing channel demonstrates that the trained HP-1 neural decoder is robust against moderate environmental noise. Parameterizing the noise strength by η , the held-out recovery accuracy remains stable up to $\eta \approx 0.215$. Performance degrades at higher noise levels, dropping to 0.491 for 9 qubits and 0.881 for 10 qubits at $\eta = 0.464$. At the fully depolarizing limit ($\eta = 1$), the subgroup information is erased, reducing the model to chance-level guessing. Appendix ?? provides detailed settings, complete numerical results, and corresponding interpretations.

Summary and outlook.— We propose a hardware-

efficient framework for solving the HSP by shallow, shift-invariant HP-1 circuits with a polynomial-scaling neural network. To verify the scalability of this protocol, we analyzed the growth model of the DFI. Finally, we evaluated our protocol on Shor’s algorithm. The numerical simulations demonstrate that our approach successfully recovers the hidden subgroup with high accuracy, while remaining robust against moderate depolarizing noise.

Realizing a practical HSP protocol requires an optimized oracle U_f implementations. Current strategies to mitigate the cost of U_f include $\mathcal{O}(1)$ -query classical pre-training of diagonal Hamiltonians [?], space-time trade-offs for non-Clifford gate reductions [?], and automatic oracle synthesis frameworks [?].

Apart from the $\mathcal{O}(n)$ scaling being hardware friendly, the grouped phase gates in HPCircuits reduces are easier to achieve compared to individual phase gate execution. This hardware-level efficiency arises because consecutive phase operations can be merged into single continuous pulses across multiple experimental platforms, including in linear optics [? ?], neutral-atom systems [? ? ?], trapped ions [? ?], and NMR [?].

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APPENDIX

A. Hidden Subgroup Problem and Fourier Sampling

In this section, we outline the standard method for solving the Hidden Subgroup Problem (HSP) over finite abelian groups [? ? ?]. Formally, the HSP is defined as follows:

Definition 3 (Hidden Subgroup Problem). *Given a function $f : G \rightarrow X$ defined on a group G , with the promise that there exists a hidden subgroup $V \leq G$ such that for all $g_1, g_2 \in G$*

$$f(g_1) = f(g_2) \iff g_1 - g_2 \in V \quad (\text{A1})$$

The objective of this task is to identify the hidden subgroup V .

In the context of this paper, we restrict our attention to finite abelian groups. For such groups, the fundamental theorem of finite abelian groups allows us to express the group structure as a direct product of cyclic groups:

$$G \cong \mathbb{Z}_{p_1} \times \mathbb{Z}_{p_2} \times \cdots \times \mathbb{Z}_{p_k}. \quad (\text{A2})$$

The standard quantum procedure for solving the hidden subgroup problem begins by preparing a uniform superposition over all elements of the group G [? ? ?]. We initialize the system in the state

$$|\psi\rangle = \frac{1}{\sqrt{|G|}} \sum_{g \in G} |g\rangle |0\rangle, \quad (\text{A3})$$

where $|G|$ is the element num of the group G . The first register holds the group elements and the second register is initialized to a standard basis state. Next, we evaluate the quantum oracle corresponding to the function f , which acts on the two registers and entangles the group elements with their corresponding function values

$$|\psi\rangle = \frac{1}{\sqrt{|G|}} \sum_{g \in G} |g\rangle |f(g)\rangle. \quad (\text{A4})$$

Since the function f is constant on the cosets of the hidden subgroup V and distinct on different cosets, it is highly advantageous to group the terms in the superposition according to these cosets.

Before proceeding, it is useful to formalize the state notation for a subset. For any subset $S \subseteq G$, we define its corresponding normalized quantum state $|S\rangle$ as the uniform superposition over all elements within the set:

$$|S\rangle = \frac{1}{\sqrt{|S|}} \sum_{s \in S} |s\rangle. \quad (\text{A5})$$

Using this definition, the normalized state for a specific coset $c + V$ is directly given by

$$|c + V\rangle = \frac{1}{\sqrt{|V|}} \sum_{v \in V} |c + v\rangle. \quad (\text{A6})$$

By applying this coset state definition, the post-oracle state can be elegantly rewritten as a superposition of these orthogonal coset states, each perfectly entangled with its unique function evaluation:

$$|\psi\rangle = \sqrt{\frac{|V|}{|G|}} \sum_{c \in C} |c + V\rangle |f(c)\rangle, \quad (\text{A7})$$

where C is the set of all possible coset representatives. For simplicity, we assume that the second register is measured immediately after the oracle query. Measuring the second register yields a specific outcome $f(c)$ with a uniform probability of $|V|/|G|$. Consequently, the first register collapses into the corresponding pure coset state:

$$|\psi_{\text{post}}\rangle = |c + V\rangle. \quad (\text{A8})$$

From this point forward, our analysis will focus exclusively on the quantum state of this first register.

These two descriptions are different at the state level, but they induce the same Fourier-basis statistics because every coset state has the same measurement distribution after the quantum Fourier transform.

A direct measurement in the computational basis is uninformative. Indeed, using the definition of the coset state,

$$\langle g | c + V \rangle = \frac{1}{\sqrt{|V|}} \sum_{v \in V} \langle g | c + v \rangle = \frac{1}{\sqrt{|V|}} \sum_{v \in V} \delta_{g, c+v}. \quad (\text{A9})$$

For fixed c , the map $v \mapsto c + v$ is a bijection from V to the coset $c + V$. Therefore the sum $\sum_{v \in V} \delta_{g, c+v}$ equals 1 when $g \in c + V$ and equals 0 when $g \notin c + V$. Hence

$$\Pr[g | c, V] = |\langle g | c + V \rangle|^2 = \begin{cases} 1/|V|, & g \in c + V, \\ 0, & g \notin c + V, \end{cases} \quad (\text{A10})$$

so the observed element is uniform on the unknown coset. Averaging over uniformly random cosets gives the marginal distribution

$$\Pr[g] = \frac{1}{|G|}, \quad (\text{A11})$$

which contains no direct information about the hidden subgroup V .

To extract meaningful information, we must apply the quantum Fourier transform (QFT) over the group G before measuring [? ?]. Let $\chi_y(g)$ denote the group characters. The QFT maps a computational basis state $|g\rangle$ as follows:

$$\text{QFT} |g\rangle = \frac{1}{\sqrt{|G|}} \sum_{y \in G} \chi_y(g) |y\rangle. \quad (\text{A12})$$

Applying this transformation to our pure coset state $|c + V\rangle$ yields:

$$\begin{aligned} \text{QFT} |c + V\rangle &= \frac{1}{\sqrt{|V|}} \sum_{v \in V} \text{QFT} |c + v\rangle \\ &= \frac{1}{\sqrt{|V||G|}} \sum_{v \in V} \sum_{y \in G} \chi_y(c + v) |y\rangle \\ &= \frac{1}{\sqrt{|V||G|}} \sum_{y \in G} \chi_y(c) \left(\sum_{v \in V} \chi_y(v) \right) |y\rangle. \end{aligned} \quad (\text{A13})$$

By the orthogonality of group characters, the sum $\sum_{v \in V} \chi_y(v)$ evaluates to $|V|$ if the character χ_y is trivial on V (i.e., y belongs to the orthogonal subgroup $V^\perp$), and evaluates to 0 otherwise. Thus, the state simplifies to a superposition over $V^\perp$:

$$\text{QFT} |c + V\rangle = \sqrt{\frac{|V|}{|G|}} \sum_{y \in V^\perp} \chi_y(c) |y\rangle. \quad (\text{A14})$$

Measuring this state in the computational basis gives the probability of observing an outcome $y \in V^\perp$:

$$\Pr[y] = \left| \sqrt{\frac{|V|}{|G|}} \chi_y(c) \right|^2 = \frac{|V|}{|G|} |\chi_y(c)|^2 = \frac{|V|}{|G|}. \quad (\text{A15})$$

Crucially, because $|\chi_y(c)|^2 = 1$ for all group characters, the resulting measurement distribution is uniformly distributed over $V^\perp$ and is strictly independent of the random coset representative c . This shift invariance is the fundamental property that allows the extraction of subgroup information by effectively removing the unknown shift at the probability level.

The quantum Fourier transform over G is defined using the character group $\hat{G}$ by

$$F_G|g\rangle = \frac{1}{\sqrt{|G|}} \sum_{\chi \in \hat{G}} \chi(g) |\chi\rangle, \quad (\text{A16})$$

where $\{|\chi\rangle : \chi \in \hat{G}\}$ is the Fourier basis and each character $\chi : G \rightarrow \mathbb{C}^\times$ is a group homomorphism. Here $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$ denotes the multiplicative group of nonzero complex numbers, so with additive notation on G one has $\chi(g_1 + g_2) = \chi(g_1)\chi(g_2)$. For finite abelian G , every character has unit modulus. Applying F_G to a coset state gives

$$\begin{aligned} F_G|c + V\rangle &= \frac{1}{\sqrt{|V|}} \sum_{v \in V} F_G|c + v\rangle \\ &= \frac{1}{\sqrt{|V||G|}} \sum_{\chi \in \hat{G}} \sum_{v \in V} \chi(c + v) |\chi\rangle \\ &= \frac{1}{\sqrt{|V||G|}} \sum_{\chi \in \hat{G}} \chi(c) \left(\sum_{v \in V} \chi(v) \right) |\chi\rangle. \end{aligned} \quad (\text{A17})$$

The character sum over V satisfies

$$\sum_{v \in V} \chi(v) = \begin{cases} |V|, & \chi \in V^\perp, \\ 0, & \chi \notin V^\perp, \end{cases} \quad V^\perp = \{\chi \in \hat{G} : \chi(v) = 1 \text{ for all } v \in V\}. \quad (\text{A18})$$

Hence

$$F_G|c + V\rangle = \sqrt{\frac{|V|}{|G|}} \sum_{\chi \in V^\perp} \chi(c) |\chi\rangle. \quad (\text{A19})$$

The Fourier-basis measurement distribution is therefore

$$p(\chi | c, V) = \begin{cases} |V|/|G|, & \chi \in V^\perp, \\ 0, & \chi \notin V^\perp, \end{cases} \quad (\text{A20})$$

which is independent of the coset representative c . The same formula holds for the mixed state ρ_V , since it is a convex combination of pure coset states with identical Fourier statistics.

Repeating the procedure therefore produces samples from $V^\perp$. In the finite abelian setting, classical post-processing reconstructs $V^\perp$ from the sample span and then recovers the hidden subgroup using the duality identity [? ? ?]

$$(V^\perp)^\perp = V. \quad (\text{A21})$$

Fourier sampling thus converts hidden translational structure in the oracle into explicit linear constraints in the dual group.

B. Necessary condition of shift invariance

Here we make explicit the probability-level shift-invariance condition used in Fourier sampling. Let G be a finite abelian group with $N = |G|$, and let $\{|g\rangle : g \in G\}$ denote the computational basis. For $g \in G$, define the shift operator

$$T_g |h\rangle = |h + g\rangle. \quad (\text{B1})$$

For a subgroup $V \leq G$, write the normalized subgroup state as

$$|V\rangle = \frac{1}{\sqrt{|V|}} \sum_{v \in V} |v\rangle. \quad (\text{B2})$$

The shift-invariance requirement relevant to HSP sampling is that the measurement distribution after applying U is independent of the coset representative:

$$|\langle x|U|V\rangle|^2 = |\langle x|UT_g|V\rangle|^2, \quad \forall x, g \in G, \forall V \leq G. \quad (\text{B3})$$

Thus $U|V\rangle$ and $U|g+V\rangle$ may differ by phases, but they must produce the same probabilities in the computational basis.

This is the operational notion of shift invariance relevant for HSP sampling. The random coset representative should disappear from the observed distribution after the transformation, even though it need not disappear from the output state itself. Any candidate replacement for the QFT must therefore satisfy Eq. (??) if it is to reproduce the same basic sampling mechanism.

The next question is how restrictive Eq. (??) already is at the level of matrix elements. A useful simplification is that one can extract a necessary condition from the trivial subgroup $V = \{0\}$ alone. In that case the subgroup state is just a computational-basis state, so the shift-invariance requirement becomes a direct constraint on how the columns of U are seen from each measurement outcome x .

The idea is simple. If the output probabilities for $|0\rangle$ and $|g\rangle$ must agree for every shift g , then for each fixed x all entries in the x -th row of U must have the same modulus. Unitarity then forces that common modulus to be exactly $1/\sqrt{|G|}$. This is the content of Theorem ??, which we restate here in the precise form used in this appendix.

Theorem (Necessary modulus condition). *Suppose U is a unitary on the group register that satisfies Eq. (??). Then, for every $x, y \in G$,*

$$|\langle x|U|y\rangle| = \frac{1}{\sqrt{|G|}}. \quad (\text{B4})$$

Equivalently, there exist phases $\theta_{xy} \in \mathbb{R}$ such that

$$\langle x|U|y\rangle = \frac{1}{\sqrt{|G|}} e^{i\theta_{xy}}. \quad (\text{B5})$$

Proof of Theorem ??. Take the trivial subgroup $V = \{0\}$. Then $|V\rangle = |0\rangle$ and $T_g|V\rangle = |g\rangle$ for every $g \in G$. Equation (??) therefore implies

$$|\langle x|U|0\rangle|^2 = |\langle x|U|g\rangle|^2, \quad \forall x, g \in G. \quad (\text{B6})$$

With the convention $U_{xy} = \langle x|U|y\rangle$, this says that for each fixed output label x , all entries in the x -th row of U have the same squared modulus. Denote this common value by a_x . Since U is unitary, each row is normalized:

$$1 = \sum_{y \in G} |U_{xy}|^2 = Na_x. \quad (\text{B7})$$

Hence $a_x = 1/N$ for every $x \in G$. Therefore every matrix element satisfies

$$|U_{xy}| = \frac{1}{\sqrt{N}} = \frac{1}{\sqrt{|G|}}. \quad (\text{B8})$$

Because this modulus is nonzero, each entry admits a polar decomposition

$$U_{xy} = \frac{1}{\sqrt{|G|}} e^{i\theta_{xy}}, \quad \theta_{xy} \in \mathbb{R}. \quad (\text{B9})$$

This proves the claim. □

C. HP- k circuits are shift invariant over $\mathbb{Z}_{2^n}$

This appendix aims to rigorously prove that the HP- k circuit family natively satisfies the shift invariant property. Establishing this theorem holds significant practical importance: it fundamentally guarantees that the output statistics obtained from this shallow circuit structure can be reliably used for the classical post-processing reconstruction of the hidden subgroup.

Theorem. *The HP- k circuit is shift-invariant over $\mathbb{Z}_{2^n}$.*

The core strategy of the proof is as follows: First, we explicitly calculate the sequential effect of each layer of the HP- k circuit on the initial basis state, deriving the analytical expression of the final quantum state. Next, we obtain the corresponding probability amplitudes by calculating the matrix elements of this output state in the computational basis. Finally, we compute the squared modulus of these amplitudes over the subgroup V and its coset $c + V$ respectively, using algebraic simplification to factor out a pure phase factor related to the coset representative c . Because the modulus of this pure phase factor is strictly 1, it naturally cancels out when calculating the measurement probabilities, thereby rigorously proving mathematically that the output probability distribution is completely independent of random shifts.

Proof. The first step is to calculate the effect of the first layer, explicitly keeping the normalization factor $\frac{1}{\sqrt{2}}$ from the Hadamard gates:

$$\begin{aligned} & \text{CP}_{\Lambda_1, \bar{\Lambda}_1}(\boldsymbol{\theta}_{\Lambda_1, \bar{\Lambda}_1}) \bigotimes_{i \in \Lambda_1} H_i |b_1 b_2 \cdots b_n\rangle \\ &= \text{CP}_{\Lambda_1, \bar{\Lambda}_1}(\boldsymbol{\theta}_{\Lambda_1, \bar{\Lambda}_1}) \bigotimes_{i \in \Lambda_1} \frac{1}{\sqrt{2}}(|0\rangle + e^{\pi i b_i} |1\rangle) \bigotimes_{k \in \bar{\Lambda}_1} |b_k\rangle \end{aligned} \quad (\text{C1})$$

$$= \bigotimes_{i \in \Lambda_1} \frac{1}{\sqrt{2}} \left(|0\rangle + \exp \left[\pi i b_i + i \sum_{j \in \bar{\Lambda}_1} \theta_{ij} b_j \right] |1\rangle \right) \bigotimes_{k \in \bar{\Lambda}_1} |b_k\rangle. \quad (\text{C2})$$

Note that since the qubits in $\bar{\Lambda}_1$ have not yet been acted upon by Hadamard gates, they remain in the computational basis.

Since the control subsets form a partition of the qubits:

$$\bigcup_{l=1}^L \Lambda_l = \{1, 2, \dots, n\}, \quad \Lambda_i \cap \Lambda_j = \emptyset, \quad (\text{C3})$$

and the target subsets are defined as the remaining unused qubits:

$$\bar{\Lambda}_l = \{1, 2, \dots, n\} \setminus \left(\bigcup_{k=1}^l \Lambda_k \right), \quad (\text{C4})$$

the sequential action of all L layers ensures every qubit i acts as a target exactly once. Let $l(i)$ denote the specific stage where $i \in \Lambda_{l(i)}$. The final state after the full circuit becomes:

$$U_{\text{HP}}^{(L)}(\boldsymbol{\theta}) |b_1 b_2 \cdots b_n\rangle = \bigotimes_{i=1}^n \frac{1}{\sqrt{2}} \left(|0\rangle + \exp \left[\pi i b_i + i \sum_{j \in \bar{\Lambda}_{l(i)}} \theta_{ij} b_j \right] |1\rangle \right). \quad (\text{C5})$$

To find the matrix element, we take the inner product with $\langle b'_1 b'_2 \cdots b'_n |$. For each qubit i , the $|1\rangle$ component is selected if and only if $b'_i = 1$, which mathematically corresponds to multiplying the entire phase exponent by b'_i . Thus, we obtain:

$$\begin{aligned} & \langle b'_1 b'_2 \cdots b'_n | U_{\text{HP}}^{(L)}(\boldsymbol{\theta}) |b_1 b_2 \cdots b_n\rangle \\ &= \frac{1}{\sqrt{2^n}} \prod_{i=1}^n \exp \left[\pi i b_i b'_i + i \sum_{j \in \bar{\Lambda}_{l(i)}} \theta_{ij} b'_i b_j \right] \end{aligned} \quad (\text{C6})$$

$$= \frac{1}{\sqrt{2^n}} \exp \left[i\pi \sum_{i=1}^n b_i b'_i + i \sum_{l=1}^{L-1} \sum_{i \in \Lambda_l} \sum_{j \in \bar{\Lambda}_l} \theta_{ij} b'_i b_j \right]. \quad (\text{C7})$$

To simplify the notation for the matrix elements while respecting the sequential asymmetry of the circuit (where each qubit pair interacts only once), we define a directed effective phase matrix $\tilde{\Theta}$ whose elements are given by:

$$\tilde{\theta}_{ij} = \begin{cases} \theta_{ij}, & \text{if } i \in \Lambda_{l(i)}, j \in \bar{\Lambda}_{l(i)} \text{ (i.e., } l(i) < l(j)) \\ \pi, & \text{if } i = j \\ 0, & \text{otherwise (i.e., } l(i) \geq l(j)). \end{cases} \quad (\text{C8})$$

Using this effective matrix, the exponent can be compactly written as a full summation over all i, j :

$$\langle b'_1 b'_2 \cdots b'_n | U_{\text{HP}}^{(L)}(\boldsymbol{\theta}) | b_1 b_2 \cdots b_n \rangle = \frac{1}{\sqrt{2^n}} \exp \left[i \sum_{i,j} \tilde{\theta}_{ij} b'_i b_j \right]. \quad (\text{C9})$$

The shift invariance requires that the probability distribution over the subgroup V is identical to its cosets. First, we compute the squared probability amplitude over V :

$$\left| \langle x | U_{\text{HP}}^{(L)}(\boldsymbol{\theta}) | V \rangle \right|^2 = \frac{1}{2^n |V|} \left| \sum_{y \in V} \exp \left[i \sum_{i,j} \tilde{\theta}_{ij} x_i y_j \right] \right|^2. \quad (\text{C10})$$

Since $V \leq \mathbb{Z}_{2^n}$, $V = \{q2^s \mid q \in \mathbb{Z}\}$ for certain $s \leq n$. Assuming a standard binary encoding where the s least significant bits correspond to the trailing qubits, $y \in V$ can be denoted as $y = y_1 y_2 \cdots y_{n-s} 0 \cdots 0$, with $y_i \in \{0, 1\}$. Similarly, the coset $c + V = \{q2^s + c \mid 0 \leq c < 2^s, q \in \mathbb{Z}\}$, and its elements $y \in c + V$ can be denoted by matching the trailing bits to the binary representation of c , yielding $y = y_1 y_2 \cdots y_{n-s} c_{n-s+1} \cdots c_n$, with $y_i, c_k \in \{0, 1\}$.

Therefore, we can expand the squared amplitude for the subgroup V :

$$\begin{aligned} & \left| \langle x | U_{\text{HP}}^{(L)}(\boldsymbol{\theta}) | V \rangle \right|^2 \\ &= \frac{1}{2^n |V|} \left| \sum_{y \in V} \exp \left[i \sum_{i=1}^n \sum_{j=1}^n \tilde{\theta}_{ij} x_i y_j \right] \right|^2 \end{aligned} \quad (\text{C11})$$

$$= \frac{1}{2^n 2^{n-s}} \left| \sum_{y_1, \dots, y_{n-s} \in \{0,1\}} \exp \left[i \sum_{i=1}^n \sum_{j=1}^{n-s} \tilde{\theta}_{ij} x_i y_j \right] \right|^2. \quad (\text{C12})$$

Evaluating the same expression for the coset $c + V$, we factor out the terms involving the constant bits c_k :

$$\begin{aligned} & \left| \langle x | U_{\text{HP}}^{(L)}(\boldsymbol{\theta}) | c + V \rangle \right|^2 \\ &= \frac{1}{2^n |c + V|} \left| \sum_{y \in c + V} \exp \left[i \sum_{i=1}^n \sum_{j=1}^n \tilde{\theta}_{ij} x_i y_j \right] \right|^2 \end{aligned} \quad (\text{C13})$$

$$= \frac{1}{2^n 2^{n-s}} \left| \exp \left[i \sum_{i=1}^n \sum_{k=n-s+1}^n \tilde{\theta}_{ik} x_i c_k \right] \sum_{y_1, \dots, y_{n-s} \in \{0,1\}} \exp \left[i \sum_{i=1}^n \sum_{j=1}^{n-s} \tilde{\theta}_{ij} x_i y_j \right] \right|^2 \quad (\text{C14})$$

$$= \frac{1}{2^n 2^{n-s}} \left| \exp \left[i \sum_{i=1}^n \sum_{k=n-s+1}^n \tilde{\theta}_{ik} x_i c_k \right] \right|^2 \left| \sum_{y_1, \dots, y_{n-s} \in \{0,1\}} \exp \left[i \sum_{i=1}^n \sum_{j=1}^{n-s} \tilde{\theta}_{ij} x_i y_j \right] \right|^2. \quad (\text{C15})$$

Because the modulus of the extracted pure phase factor strictly equals 1, it completely cancels out, yielding:

$$\left| \langle x | U_{\text{HP}}^{(L)}(\boldsymbol{\theta}) | c + V \rangle \right|^2 = \frac{1}{2^n 2^{n-s}} \left| \sum_{y_1, \dots, y_{n-s} \in \{0,1\}} \exp \left[i \sum_{i=1}^n \sum_{j=1}^{n-s} \tilde{\theta}_{ij} x_i y_j \right] \right|^2 \quad (\text{C16})$$

$$= \left| \langle x | U_{\text{HP}}^{(L)}(\boldsymbol{\theta}) | V \rangle \right|^2. \quad (\text{C17})$$

This completes the proof. $\square$

D. Transversal Hadamard sampling and efficiency guarantee

We now give the calculation underlying the HP-0 discussion in the main text. Consider $G = \mathbb{Z}_{2^n}$ represented by n computational-basis qubits, and use the bit ordering in which a multiple of 2^p has the form

$$h = h_1 h_2 \cdots h_{n-p} 0 \cdots 0. \quad (\text{D1})$$

The subgroup generated by 2^p is

$$H_p = \{q2^p : q = 0, \dots, 2^{n-p} - 1\}, \quad (\text{D2})$$

so $|H_p| = 2^{n-p}$. Since adding an element of H_p can vary the first $n-p$ bits, each coset has a representative of the form

$$c = 0 \cdots 0 c_{n-p+1} \cdots c_n. \quad (\text{D3})$$

Thus a normalized coset state can be written as

$$|c + H_p\rangle = \frac{1}{\sqrt{2^{n-p}}} \sum_{h_1, \dots, h_{n-p} \in \{0,1\}} |h_1 \cdots h_{n-p} c_{n-p+1} \cdots c_n\rangle. \quad (\text{D4})$$

For an n -bit string x , the transversal Hadamard transform satisfies

$$H^{\otimes n} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{k \in \{0,1\}^n} (-1)^{k \cdot x} |k\rangle. \quad (\text{D5})$$

Applying this to Eq. (??) gives

$$\begin{aligned} H^{\otimes n} |c + H_p\rangle &= \frac{1}{\sqrt{2^{n-p}}} \frac{1}{\sqrt{2^n}} \sum_{k \in \{0,1\}^n} \sum_{h_1, \dots, h_{n-p} \in \{0,1\}} (-1)^{\sum_{r=1}^{n-p} k_r h_r + \sum_{r=n-p+1}^n k_r c_r} |k\rangle \\ &= \frac{1}{\sqrt{2^{n-p}}} \frac{1}{\sqrt{2^n}} \sum_{k \in \{0,1\}^n} (-1)^{\sum_{r=n-p+1}^n k_r c_r} \prod_{r=1}^{n-p} \left(\sum_{h_r \in \{0,1\}} (-1)^{k_r h_r} \right) |k\rangle. \end{aligned} \quad (\text{D6})$$

The inner product factor is zero unless $k_1 = \cdots = k_{n-p} = 0$, in which case it equals 2^{n-p} . Therefore

$$H^{\otimes n} |c + H_p\rangle = \frac{1}{\sqrt{2^p}} \sum_{k_{n-p+1}, \dots, k_n \in \{0,1\}} (-1)^{\sum_{r=n-p+1}^n k_r c_r} |0\rangle^{\otimes (n-p)} |k_{n-p+1} \cdots k_n\rangle. \quad (\text{D7})$$

The coset representative c only contributes the phase factor in Eq. (??). Consequently, the measurement distribution is independent of c : the first $n-p$ bits are always zero, while the remaining p bits are uniformly random.

This immediately gives a simple reconstruction procedure for the idealized $\mathbb{Z}_{2^n}$ HSP. Each application of the oracle-and-measurement routine prepares a random coset state, and Eq. (??) shows that a subsequent transversal Hadamard layer converts that coset state into a sample whose first $n-p$ bits are deterministically zero while the remaining p bits are unbiased. The resulting sampling-and-recovery routine is summarized in Algorithm ??.

Algorithm 1: Transversal Hadamard Sampling for $H_p \leq \mathbb{Z}_{2^n}$

Input: Oracle access to an HSP instance over $\mathbb{Z}_{2^n}$, sample count m

Output: Hidden subgroup $H_p = \langle 2^p \rangle$

```

1 Initialize an empty list of samples  $\mathcal{B}$ ;
2 for  $i = 1$  to  $m$  do
3   Prepare the uniform superposition  $\frac{1}{\sqrt{2^n}} \sum_{g \in \mathbb{Z}_{2^n}} |g\rangle |0\rangle$ ;
4   Query the oracle and measure the second register to obtain a coset state  $|c_i + H_p\rangle$ ;
5   Apply  $H^{\otimes n}$  to the first register;
6   Measure the first register and record the bit string  $b^{(i)} \in \{0, 1\}^n$ ;
7   Append  $b^{(i)}$  to  $\mathcal{B}$ ;
8 end
9 for  $j = 1$  to  $n$  do
10  Mark bit position  $j$  as fixed if  $b_j^{(i)} = 0$  for all recorded samples;
11  Otherwise mark bit position  $j$  as free;
12 end
13 Let  $p$  be the number of free trailing bit positions;
14 return  $H_p = \langle 2^p \rangle$ ;

```

Algorithm ?? is correct because, under the bit-ordering convention above, the free positions occur exactly in the final p bits. Equivalently, the measurement data have the form

$$0 \cdots 0 * \cdots *, \quad (\text{D8})$$

where each star denotes an unbiased bit. Identifying which positions ever take the value 1 therefore determines p , and hence determines the subgroup $H_p = \langle 2^p \rangle$.

The sample complexity is logarithmic in the number of qubits. For Algorithm ??, fixed bits are never misclassified, because they are exactly zero in every sample. For a free bit, the probability that it appears as zero in all m independent samples is 2^{-m} . By the union bound, the probability of misclassifying at least one free bit is at most

$$p2^{-m} \leq n2^{-m}. \quad (\text{D9})$$

Thus $m \geq \lceil \log_2(n/\epsilon) \rceil$ samples are sufficient for Algorithm ?? to identify the fixed and free bit positions, and therefore the hidden subgroup H_p , with probability at least $1 - \epsilon$.

E. QFT Fisher information for subgroup-generator estimation

To benchmark the proposed circuit family, we treat the hidden subgroup generator r as the parameter to be inferred from the measurement distribution produced by the standard QFT. Consider the truncated periodic state

$$|\psi_r\rangle = \frac{1}{\sqrt{R}} \sum_{q=0}^{R-1} |qr\rangle, \quad (\text{E1})$$

where R is the number of multiples of r supported in the 2^n -dimensional register. For the 2^n -point QFT,

$$|j\rangle \mapsto \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi i j k / 2^n} |k\rangle, \quad (\text{E2})$$

and therefore

$$|\psi_r\rangle \mapsto \frac{1}{\sqrt{R2^n}} \sum_{k=0}^{2^n-1} \left(\sum_{q=0}^{R-1} e^{2\pi i q r k / 2^n} \right) |k\rangle. \quad (\text{E3})$$

The corresponding measurement probability is

$$p(k|r) = \frac{1}{R2^n} \left| \sum_{q=0}^{R-1} e^{2\pi i q r k / 2^n} \right|^2 = \frac{1}{R2^n} \left(\frac{\sin(\pi R r k / 2^n)}{\sin(\pi r k / 2^n)} \right)^2. \quad (\text{E4})$$

In the regime $r \ll 2^n$, it is convenient to introduce the continuum variable $x = \pi k/2^n \in [0, \pi]$ and approximate the distribution by the density

$$f(x|r) = \frac{1}{\pi R} \left(\frac{\sin(Rrx)}{\sin(rx)} \right)^2. \quad (\text{E5})$$

The QFT Fisher information with respect to the generator r is then

$$I_{\text{QFT}}(r) = \int_0^\pi f(x|r) \left[\frac{\partial}{\partial r} \ln f(x|r) \right]^2 dx, \quad (\text{E6})$$

with score function

$$\frac{\partial}{\partial r} \ln f(x|r) = 2Rx \cot(Rrx) - 2x \cot(rx). \quad (\text{E7})$$

The integral representation above makes the statistical meaning of the benchmark transparent, but it still hides the dependence on the support size R inside a strongly oscillatory quotient of sines. For the comparison carried out in the main text, what we need is a formula in which the explicit R -scaling is separated from the oscillatory corrections.

There is one additional simplification that closes this gap. If we write

$$S(u) = \frac{\sin(Ru)}{\sin u}, \quad (\text{E8})$$

then $f(x|r) = S(rx)^2/(\pi R)$, and the score function implies

$$f(x|r) \left[\frac{\partial}{\partial r} \ln f(x|r) \right]^2 = \frac{4x^2}{\pi R} [S'(rx)]^2. \quad (\text{E9})$$

Thus the Fisher-information integral is reduced to the square of a finite trigonometric polynomial. Once that polynomial is expanded into cosine modes, the remaining calculation involves only the moments of x^2 against those modes.

Theorem 2 (QFT Fisher-information benchmark). *For the continuum density*

$$f(x|r) = \frac{1}{\pi R} \left(\frac{\sin(Rrx)}{\sin(rx)} \right)^2, \quad x \in [0, \pi], \quad (\text{E10})$$

the Fisher information with respect to the generator r is

$$I_{\text{QFT}}(r) = \frac{4\pi^2}{9}(R^2 - 1) + \frac{8}{3\pi R} \sum_{j=1}^{R-1} (R-j)(R^2 - 2Rj - 2j^2 - 1) M_j(r),$$

$$M_j(r) = \frac{4\pi jr \cos(2\pi jr) + (4\pi^2 j^2 r^2 - 2) \sin(2\pi jr)}{(2jr)^3}. \quad (\text{E11})$$

In particular, since $M_j(r) = O(r^{-2})$ for fixed j , the large- r regime is governed by the non-oscillatory term

$$I_{\text{QFT}}(r) \approx \frac{4\pi^2}{9}(R^2 - 1). \quad (\text{E12})$$

To derive Theorem ??, it is convenient to pass from the Fejer-kernel representation of $S(u)^2$ to an exponential representation of $S(u)$ itself. Differentiating that exponential sum yields a finite trigonometric polynomial, and squaring it produces an exact cosine expansion for $[S'(u)]^2$. After that step, the Fisher-information integral reduces to $\int_0^\pi x^2 dx$ and $\int_0^\pi x^2 \cos(2jrx) dx$. The asymptotic formula follows because the oscillatory moments decay as r^{-2} .

Proof. From

$$S(u) = \frac{\sin(Ru)}{\sin u} = \sum_{m=0}^{R-1} e^{i(R-1-2m)u}, \quad (\text{E13})$$

we obtain

$$S'(u) = i \sum_{m=0}^{R-1} (R-1-2m) e^{i(R-1-2m)u}. \quad (\text{E14})$$

Squaring this expression and grouping equal frequencies gives an even cosine series

$$[S'(u)]^2 = A_0 + 2 \sum_{j=1}^{R-1} A_j \cos(2ju), \quad (\text{E15})$$

where

$$A_j = - \sum_{m=0}^{R-1-j} (R-1-2m)(2j-R+1+2m) = \frac{(R-j)(R^2-2Rj-2j^2-1)}{3}, \quad j = 0, \dots, R-1. \quad (\text{E16})$$

In particular,

$$A_0 = \frac{R(R^2-1)}{3}. \quad (\text{E17})$$

Since

$$I_{\text{QFT}}(r) = \frac{4}{\pi R} \int_0^\pi x^2 [S'(rx)]^2 dx, \quad (\text{E18})$$

substituting the cosine expansion yields

$$I_{\text{QFT}}(r) = \frac{4}{\pi R} \left[A_0 \int_0^\pi x^2 dx + 2 \sum_{j=1}^{R-1} A_j \int_0^\pi x^2 \cos(2jrx) dx \right]. \quad (\text{E19})$$

The required moments are

$$\int_0^\pi x^2 dx = \frac{\pi^3}{3} \quad (\text{E20})$$

and

$$\int_0^\pi x^2 \cos(2jrx) dx = \frac{4\pi jr \cos(2\pi jr) + (4\pi^2 j^2 r^2 - 2) \sin(2\pi jr)}{(2jr)^3}. \quad (\text{E21})$$

Substituting these expressions gives

$$\begin{aligned} I_{\text{QFT}}(r) &= \frac{4}{\pi R} \left[\frac{R(R^2-1)}{3} \cdot \frac{\pi^3}{3} + \frac{2}{3} \sum_{j=1}^{R-1} (R-j)(R^2-2Rj-2j^2-1) M_j(r) \right] \\ &= \frac{4\pi^2}{9} (R^2-1) + \frac{8}{3\pi R} \sum_{j=1}^{R-1} (R-j)(R^2-2Rj-2j^2-1) M_j(r), \end{aligned} \quad (\text{E22})$$

which is the stated exact formula.

Finally, the oscillatory correction term is subleading because $M_j(r) = O(r^{-2})$ for fixed j . Keeping only the constant contribution therefore gives

$$I_{\text{QFT}}(r) \approx \frac{4\pi^2}{9} (R^2-1). \quad (\text{E23})$$

This proves the theorem. □

Theorem ?? provides the analytical QFT benchmark used in the main text.

F. Log-linear growth fit for the numerical Fisher-information scan

The fixed- r QFT benchmark derived above clarifies the dominant scaling mechanism, but Fig. ??(b) probes a slightly different quantity. There, the vertical axis is the minimum discrete Fisher information over the period window used in the numerical scan. For the qubit range shown in Fig. ??(b), that window extends up to $r_{\max}(n) = n^2$, so the relevant QFT comparator is the worst case within the same scan rather than a single fixed value of r .

The leading term in Theorem ?? is monotone in R , and in the sampled regime $R \approx 2^n/r$. Consequently the smallest QFT Fisher information inside the scan window is attained near the largest sampled period, which yields the window-matched estimate

$$I_{\text{QFT,win}}(n) \approx \frac{4\pi^2}{9} \left(\left\lfloor \frac{2^n}{n^2} \right\rfloor^2 - 1 \right). \quad (\text{F1})$$

If one keeps only the dominant exponential factor, this becomes

$$\ln I_{\text{QFT,win}}(n) = (2 \ln 2) n - 4 \ln n + O(1), \quad (\text{F2})$$

so the asymptotic QFT exponent is $k_{\text{QFT},\infty} = 2 \ln 2 \approx 1.386$, while a direct log-linear fit over the finite window $n = 7, \dots, 14$ gives the effective slope $k_{\text{QFT,win}} = 1.067$.

This observation suggests a simple diagnostic for the numerical data in Fig. ??(b). If the plotted curves retain a stable exponential dependence on the qubit count over the sampled window, then the transformed data should be well approximated by

$$\ln I_{\min}(n) = kn + b. \quad (\text{F3})$$

We therefore perform ordinary least-squares fits to the log-transformed mean values of each curve shown in Fig. ??(b). Figure ?? displays the resulting regressions.

Circuit	qubits	k	95% CI	R^2
ph1	7–14	0.378	[0.306, 0.449]	0.965
optimized 1ph	7–12	0.443	[0.253, 0.632]	0.913
ph1_random	7–14	0.386	[0.318, 0.453]	0.970
ph_random	7–14	0.406	[0.361, 0.451]	0.988
ph_random_phase	7–14	0.363	[0.319, 0.407]	0.986

TABLE I. Effective exponents obtained from the log-linear model $\ln I_{\min}(n) = kn + b$ applied to the numerical Fisher-information curves in Fig. ??(b).

The fitted lines account for nearly all of the log-scale variance across the sampled window. The coefficient of determination ranges from 0.913 to 0.988, the log-space root-mean-square residual lies between 0.101 and 0.233, and every fitted slope is significantly positive with $p \leq 2.9 \times 10^{-3}$. In that sense, a linear regression on $\ln I_{\min}$ is justified as a finite-window descriptive model for these curves. At the same time, the fitted quantity should be interpreted as an effective exponent on the restricted range $n = 7, \dots, 14$, not as a claim about the ultimate asymptotic law.

The comparison with the QFT benchmark is unambiguous. Even after matching the period window used in the numerical scan, the QFT reference fit remains much steeper, with $k_{\text{QFT,win}} = 1.067$ and $R^2 = 0.9966$. The empirical circuit families therefore grow substantially more slowly: their fitted exponents occupy only 34%–41% of the window-matched QFT slope, and only 26%–32% of the asymptotic QFT exponent $2 \ln 2$. Among the plotted families, optimized 1ph has the largest point estimate $k = 0.443$, although its confidence interval is widest because that dataset currently stops at 12 qubits. Among the curves available through 14 qubits, ph_random has the largest fitted exponent, $k = 0.406$.

This comparison sharpens the interpretation of Fig. ??(b). The HP-family circuits retain a clear exponential-in- n growth trend in the minimum Fisher information over the sampled window, which is why the log-linear fit works well, but they do not reproduce the QFT growth rate. Their advantage in the present setting is therefore not that they match the full QFT benchmark asymptotically, but that they retain a nontrivial and practically usable fraction of that information while remaining much shallower.

fi_fig/fi_log_regression_plot.pdf

FIG. 4. Log-linear fits of the qubit-scaling curves in Fig. ??(b). Markers denote the mean Fisher-information values plotted in Fig. ??(b), after applying the transformation $I_{\min} \mapsto \ln I_{\min}$, and solid lines denote the corresponding least-squares fits $\ln I_{\min}(n) = kn + b$.

G. Noise-robustness experiment details

We evaluate noise robustness using the previously trained 9- and 10-qubit HP-1 plus neural-decoder pipelines from the period-recovery study. For each system size, we use a measurement budget of $1024n^2$ samples per instance, giving 82944 measurements for $n = 9$ and 102400 for $n = 10$. The candidate periods are $r \in \{9, \dots, 80\}$ for 9 qubits and $r \in \{10, \dots, 99\}$ for 10 qubits. Probability distributions are generated from the exact subgroup support of the trained HP-1 circuit, and the neural decoder checkpoint is selected using noiseless validation accuracy. The selected checkpoints occur at epoch 3 with validation top-1 accuracy 0.9959 for 9 qubits, and at epoch 10 with validation top-1 accuracy 1.0000 for 10 qubits.

The evaluation set holds out 3 shifts for each candidate period and generates 8 independent noisy redraws for each

held-out shift. This yields $72 \times 3 \times 8 = 1728$ test examples for 9 qubits and $90 \times 3 \times 8 = 2160$ test examples for 10 qubits. To probe robustness after the shallow quantum transformation, we insert a single global depolarizing channel after the full HP-1 circuit and before measurement. At the probability level, this is implemented as

$$\tilde{p}_\eta(x|r, s) = (1 - \eta)p(x|r, s) + \eta \frac{1}{2^n}, \quad (\text{G1})$$

where $p(x|r, s)$ is the ideal output distribution for period r and shift s , and η is the depolarizing strength. For each noisy distribution $\tilde{p}_\eta$, we resample $1024n^2$ outcomes and feed the resulting bit-string batch into the fixed neural decoder. The sweep uses the values

$$\eta \in \{1, 0.464, 0.215, 0.1, 0.0464, 0.0215, 0.01, 0.00464, 0.00215, 0.001\}. \quad (\text{G2})$$

η	9-qubit accuracy	10-qubit accuracy
1.000	0.0139	0.0111
0.464	0.4913	0.8810
0.215	0.9948	1.0000
0.100	0.9954	1.0000
0.0464	0.9959	1.0000
0.0215	0.9942	1.0000
0.0100	0.9954	1.0000
0.00464	0.9959	1.0000
0.00215	0.9948	1.0000
0.00100	0.9948	1.0000

TABLE II. Held-out period-recovery accuracy under post-HP-1 global depolarizing noise.

Several features of Table ?? are worth noting. First, performance is essentially unchanged throughout the weak-to-moderate noise regime and remains near the noiseless baseline up to $\eta \approx 0.215$. Second, substantial degradation appears only once the noisy distribution is strongly mixed with the uniform distribution: at $\eta = 0.464$, the accuracy drops to 0.4913 for 9 qubits and 0.8810 for 10 qubits. Finally, when $\eta = 1$, the output distribution becomes exactly uniform and is therefore independent of both the period and the shift. In that limit, all subgroup information is erased, so the best possible behavior is chance-level guessing over the candidate periods. The observed accuracies 0.0139 and 0.0111 agree exactly with the corresponding chance levels $1/72$ and $1/90$, confirming that the numerical results are consistent with the depolarizing-noise model.